HDOC DAY 57

PROBLEM

Write a C program that reads the size and elements of a variable-length integer array, arr[]. Pass the array to a function named max_subarray_sum() using its address. Use pointers to pass the array between the calling function and max_subarray_sum(). The function max_subarray_sum() must:

- Find the maximum sum of any contiguous subarray in arr[] using Kadane's algorithm.
- Display the maximum subarray sum.
- If all elements are negative, display the largest, or least negative, element.
- Ensure that the selected subarray contains at least one element.

CODE

```
#include <stdio.h>
#include <stdlib.h>

void max_subarray_sum(int *arr, int n)
{
   int max_sum = *arr;
   int current_sum = *arr;

   for (int i = 1; i < n; i++)
   {
      if (current_sum + *(arr + i) > *(arr + i))
         current_sum = current_sum + *(arr + i);
      else
         current_sum = *(arr + i);
```

```
        if (current_sum > max_sum)
            max_sum = current_sum;
    }

    printf("Maximum subarray sum = %d\n", max_sum);
}

int main()
{
    int n;

    printf("Enter the size of the array: ");
    scanf("%d", &n);

    if (n <= 0)
    {
        printf("Array size must be positive.\n");
        return 1;
    }

    int *arr = malloc(n * sizeof(int));

    if (arr == NULL)
    {
        printf("Memory allocation failed.\n");
        return 1;
    }
```

```
    printf("Enter the array elements:\n");
    for (int i = 0; i < n; i++)
    {
        scanf("%d", arr + i);
    }

    max_subarray_sum(arr, n);

    free(arr);
    return 0;
}
```